

Practice 5: XAI - Partial Dependence Plots

Andres Alvarruiz Lopez, Alejandro Legleye Enguix, Ferran Soler Martinez

10/05/2026

Contents

1	Introduction	1
2	1D PDP: Bike Rental Prediction	1
3	2D PDP: Humidity and Temperature	2
4	PDP: House Price Prediction	3
5	Conclusions	4

1 Introduction

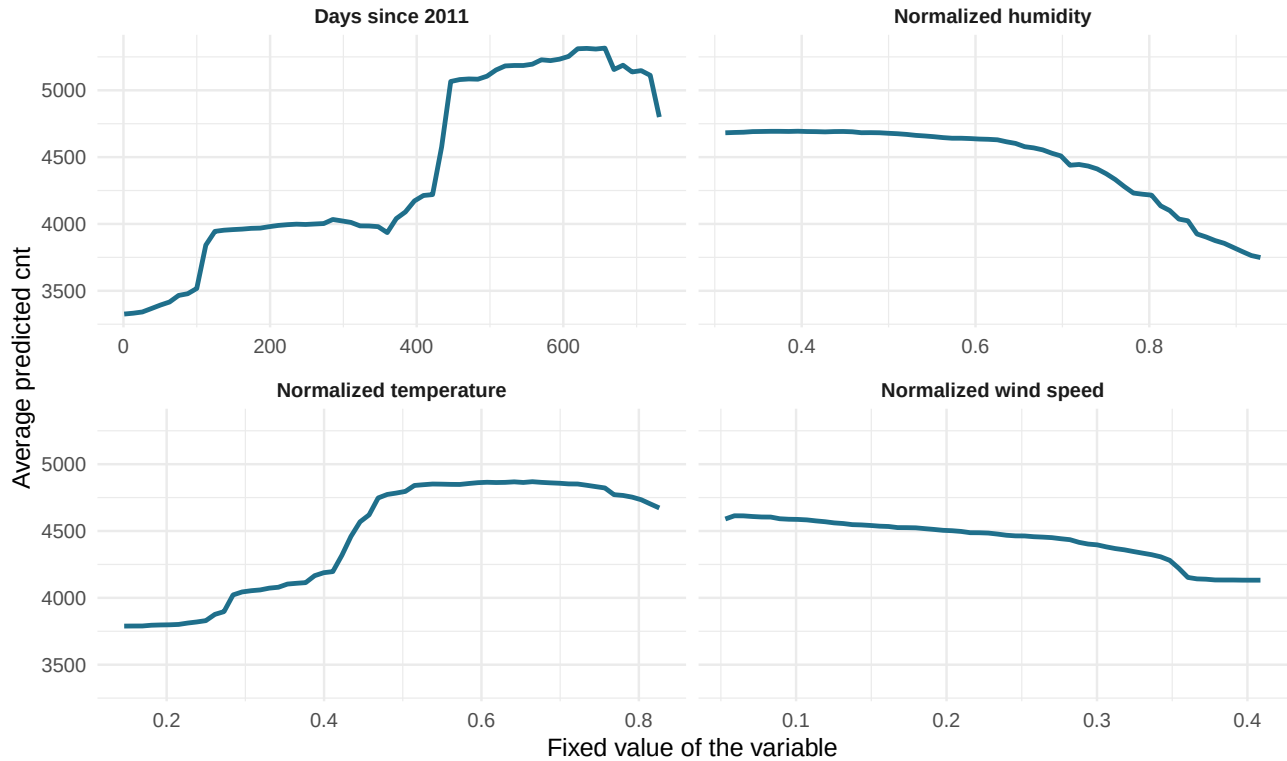
The objective of this practice is to apply model-agnostic explainability methods. Specifically, Partial Dependence Plots (PDP) are used to analyze how the average prediction of a random forest changes when one variable is forced to take different values.

A PDP does not show a causal relationship. It shows the relationship learned by the model while keeping the remaining variables distributed as observed in the data. For this reason, PDPs are useful for interpreting nonlinear models, but they should be read as explanations of the model rather than as direct laws of the real phenomenon.

2 1D PDP: Bike Rental Prediction

A regression random forest is fitted to predict `cnt`, the total number of bikes rented per day. The variables `casual` and `registered` are removed because they are direct components of `cnt` and would introduce data leakage. The approximate OOB explained variance of the model is 88.5%.

1D PDP for the bike rental model

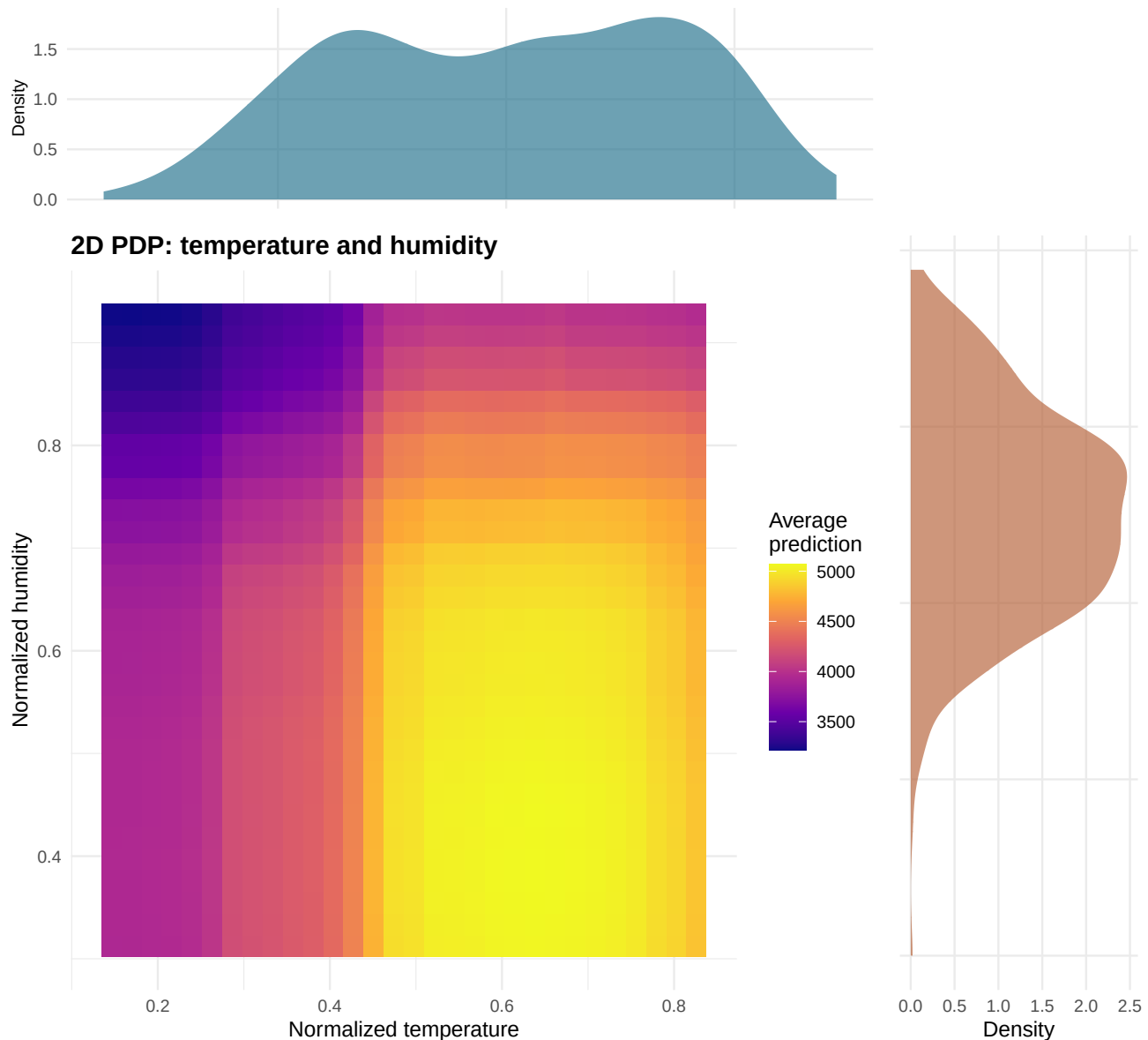


Interpretation. The number of days since 2011 has an increasing effect: the model learns a positive time trend, consistent with higher demand in the second part of the observed period. Temperature also has a clearly positive effect up to medium-high values; with more pleasant temperatures, the model predicts more bike rentals.

Humidity shows the opposite pattern: as humidity increases, especially at high levels, the average predicted number of bikes decreases. Wind speed also reduces predicted demand when it reaches high values, although its effect is weaker and more irregular than the effects of temperature or humidity.

3 2D PDP: Humidity and Temperature

For the two-dimensional PDP, a random sample of 500 observations is used, as requested in the assignment. The average prediction is computed for each combination of temperature and humidity values.



Interpretation. The highest prediction area appears with medium or high temperature and low or moderate humidity. When humidity is high, the predicted number of rentals decreases even if the temperature is favorable. This indicates that the model learns not only individual effects, but also an intuitive interaction: warm days favor bike rental, but if they are accompanied by high humidity, the expected demand decreases.

The marginal densities help identify where real observations are located. Areas of the map with low density should be interpreted more carefully, because the model is extrapolating more than in frequent regions of the data.

4 PDP: House Price Prediction

For house price prediction, a random forest is trained with the variables specified in the assignment: bedrooms, bathrooms, sqft_living, sqft_lot, floors, and yr_built. The approximate OOB ex-

plained variance of the model is 62%. To compute the PDPs, a sample of 2000 observations is used, reducing the computational cost without changing the logic of the method.



Interpretation. The variable with the clearest effect is `sqft_living`: as living area increases, the predicted price also increases. This pattern is expected because interior size is one of the most direct signals of a house’s value.

Bathrooms also increase the predicted price, although with plateau regions. This suggests that the model values the transition from few bathrooms to a medium number more than adding bathrooms when the house already has several. Bedrooms show a less linear effect: more bedrooms do not always imply a higher price if they are not accompanied by more living area or other characteristics. Floors have a moderate and irregular effect; the model seems to associate them with some increase in price, but not with a relationship as strong as living area.

5 Conclusions

PDPs make it possible to visually summarize the relationships learned by random forest models. In the bike rental case, temperature and the time trend increase the prediction, while high humidity and wind speed reduce it. In the house price case, living area dominates the explanation of price, followed by bathrooms; bedrooms and floors have more context-dependent effects.

The main precaution is that PDPs average over all observations. If variables are highly correlated or some regions have few data points, the interpretation must be made with caution.